

Dohyun (Eric) Kim

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PROFILE

Penultimate Computer Systems Engineering (Hons) student at the University of Auckland, specialising in embedded systems and digital hardware design. AWS certified in both Cloud and AI/ML, with a competition winning project portfolio spanning on FPGA development, PCB design, and embedded robotics systems.

EDUCATION

University of Auckland – Auckland CBD, New Zealand

Expected Graduation Nov 2027

- Degree: *Bachelor of Engineering (Honours) in Computer Systems Engineering*
- Concentration: Embedded Systems & Digital Hardware Design

EXPERIENCE

Robotics Instructor

Auckland, New Zealand

Creative Imaginary Lab (ciLab)

Apr 2026 - Present

- Instructed students in robot **hardware assembly and software programming** to complete assigned missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

PROJECTS

Smart Energy Monitor

Embedded C

Embedded Energy Monitoring Systems - Team Project

Oct 2025

- Designed an embedded energy monitoring system using **ATmega328P** microcontrollers with ADC processing in order to measure and display real-time energy usage.
- Produced a validated PCB schematic using **Altium Designer** in order to achieve accurate signal conditioning.

Flappy Universe

VHDL

FPGA Game Implementation

Jun 2026

- Implemented a Flappy Bird-style game in VHDL on an **Altera FPGA** with VGA output and sprite rendering in order to demonstrate real-time hardware design.
- Developed a pixel-priority **VHDL pipeline** in order to render game scenes at VGA resolution.

AWARD

3rd Place in 2025 ECSE Design Competition

Team Project

"Winnie the Bot" - Interactive companion robot

Sep 2025

- Built an AI-powered robot using dual **ATmega328P** microcontrollers, an AI camera, and audio peripherals in order to enable face tracking, arm movement, and voice dialogue in a compact form factor.
- Contributed to **prototyping** the enclosure using iterative design in order to deliver a functional physical build.

ACTIVITIES & LEADERSHIPS

Academic Team Executive

Auckland, New Zealand

KEB (Korean Engineering Body)

Jul 2024 - Present

- Developed and delivered **tutorial sessions** for junior engineering students across various courses.
- Participated in the planning and design of academic events **to support the student community**.

Official Competition Staff

NZ Robotics Olympiad 2026

IEEE R&A (IEEE Robotics and Automation Society)

Jul 2026

- Selected as official Competition Staff for NZRO 2026, drawing on prior volunteering and robotics instructor experience at ciLab.
- Supported competition operations and troubleshoot technical issues to ensure fair, smooth-running matches.

Full-time Student Volunteer

NZ Robotics Olympiad 2025

IEEE (Institute of Electrical and Electronics Engineer)

Jul 2025

- Volunteered 40+ hours supporting operations and logistics at NZRO 2025, an IEEE robotics competition.
- Collaborated with organisers to help deliver a smooth event experience for participants.

SKILLS

Languages & Frameworks: C, VHDL, Python, MATLAB, Java, R, React.js

Tools & Platforms: Altium Designer, LTSpice, ModelSim, Intel Quartus Prime, Proteus, Atmel AVR, Git, AutoCAD, AWS